

PROCESS YOUR CAREER WITH DSP!!

Signal processing, especially digital signal processing (DSP)—a highly value-added market with a bit of distinction—opens up a minefield of opportunities for aspirants

■ SUDESHNA DAS

Digital seems to be the buzzword in today's electronic experience—be it digital television, camera, phone, video or audio. Have you ever wondered why the word 'digital' is so appealing? What is the unique selling point of these products? The answer is simple—significant improvement of experience without distortion. How? The answer is—signals today are being processed digitally through digital signal processing or DSP.

"DSP has revolutionised the way we look at new electronic and telecom products in terms of size, power and efficiency. We use a myriad of devices today, all of which embed a large number of DSPs," says Manoj Misra, director and head-DSP, FPGA & ASIC Practice, Hughes Systique Corporation.

If hours and hours of signal analysis to process, amplify and appropriate interpretation attracts you then it's an industry worth considering for career development.

Catch a glimpse...

The term DSP is used to describe signal processing in digital domain either in hardware or in software. In fact, all digital devices perform signal processing—be it a mobile, radar, ECG, MRI machine, TV, camera, microwave oven, washing machine, car, satellite or an airplane—the list is very long.

India is a developing economy with embedded, IT, software industries that are growing at a modest rate of 9-11 per cent per annum. The captive needs

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and potential growth of business is luring many Indian companies/investors as well as foreign investors/companies to invest in new niche areas like telecom, communication, defence, space, medical instrumentation, consumer electronics, etc. This has resulted in many centres of excellence focusing on activities related to DSP applications. Some of the key potential areas include defence, telecom, multimedia, medical instrumentation, low-cost smart devices, space and aviation industry, consumer electronics, auto industry and safety products.

NASSCOM has projected DSP as a career with high growth prospects. DSP programming and technology is mathematics-intensive. Indian graduates may stand to have an edge over others as they tend to be skilled in number crunching. "Though there are quite a few experienced DSP engineers commanding the top positions in the industry, the gap in supply and demand of skilled DSP staff shows no sign of easing, and with ever widening applications of DSP, there is no better time to embrace and master this technology," opines C.P. Ravikumar, Texas Instruments.

The recruitment trend for this sector further confirms his statement.

Larger players like Samsung, Nokia, Motorola and Texas Instruments, as well as a host of small and medium design solution providers like Ittium, Hughes Systique and AppliedMicro, offer opportunities in DSP. Even auto companies like Ashok Leyland, Tata Motors, Mahindras, Bajaj and Maruti, and healthcare companies like Philips and Siemens offer exciting career options in signal processing. Companies like Wipro and Mindtree that invest in product development, also require DSP professionals.

What your pocket says

Let's turn to 'money matters.' It won't be an exaggeration to suggest that this field offers enough provisions for learning as well as earning. This industry is typically at the higher end of the value chain and hence the salary and rewards are at above average levels of the industry.

The pay is quite good even for freshers.

The high demand for competent staff in DSP segment creates several opportunities for aspirants. The starting salary for a fresher is typically between Rs 300,000 to Rs 500,000 per annum with impressive year to year salary hikes. As an engineer, you may choose to continue